The Hong Kong Polytechnic University

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT544
Subject Title	Artificial Intelligence and Human-Machine Interaction in Wearable Systems
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	N/A
Objectives	The subject provides fundamental knowledge of artificial intelligence (AI) and human-machine interaction (HMI) involved in wearable technologies, and also explores the latest applications of AI and HMI in wearable interactive systems. This subject is designed to introduce the interaction concepts and principles, information channels, system hardware (human, machine and sensor) and system software (intelligent information processing) in the development of wearable interactive systems, which enable students to identify the features and requirements of such systems, and also apply the skills and knowledge for the creative design of new wearable interactive systems, as well as to analyse and evaluate emerging developments in the field.
Intended Learning Outcomes *(Note 1)*	Upon completion of the subject, students will be able to: a. describe the underlying mechanisms of artificial intelligence in the advanced processing of sensing information acquired from wearable systems; b. describe the development trends of human-machine interaction and artificial intelligence and their applications in wearable interactive systems; c. describe the basic information components of human-machine interaction and human factors involved in wearable interactive systems; d. develop a comprehensive understanding of human information and its interactive process with the surrounding environment in wearable interactive systems; e. develop and apply the knowledge and skills for human-machine interaction design and evaluation;

	f. develop the ability to adapt new technologies in wearable and interactive system design and update their knowledge in the field.
Subject Synopsis/ Indicative Syllabus *(Note 2)*	**(I) Artificial Intelligence (AI) in Wearable Systems** • Sensing Information Acquisition and Processing *types of sensing information, information acquisition, popular information processing methods* • Artificial Intelligence Fundamentals *pattern analysis, machine learning, artificial neural networks, deep learning* • Applications of AI in Wearable Systems *health monitoring and diagnosis, behaviour recognition, occupational safety, smart home, etc.* **(II) Human Machine Interaction (HMI) in Wearable Systems** • Principle of an Interactive System *constituent components, interactive process* • Interaction Design *requirements analysis, user analysis, task modelling, determination of sensation/perception information, selection of sensors, interface design* • Evaluation Criterion and Evaluation Methods *usability, functionality and acceptability of an interactive system, expert evaluation, user evaluation* • Applications of HMI with Wearable Sensing Devices *fashion, retailing, healthcare, entertainment, etc.* **(III) Human Factors (HFs) in Wearable Systems** • Sensation and Perception of Information *visual, aural, tactile and haptic, multimodal information* • Human Information Processing *human problem-solving model, human reaction and prediction of cognitive performance*
Teaching/Learning Methodology *(Note 3)*	Fundamental concepts, principles, general theories, design and evaluation processes will be introduced in lectures. Applications, real cases and emerging trends of new technologies will be further discussed in tutorials, as supplementary and extension to lectures. Students will be encouraged to investigate related literature and materials, design and present their own conceptual models of new wearable interaction systems.

Assessment Methods in Alignment with Intended Learning Outcomes

(Note 4)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	f
Continuous Assessment	**50%**	✓	✓	✓	✓	✓	✓
1. Assignment	25%	✓	✓	✓	✓		✓
2. Project	25%		✓	✓	✓	✓	✓
Examination	**50%**	✓	✓	✓	✓	✓	
Total	**100 %**						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Continuous assessments including an assignment and a project will be given to assess the learning outcomes at different learning stages to enhance the understanding of the theories and develop the skills to apply the theories. The final examination will be used to evaluate the overall understanding of human factors, human-machine interaction and artificial intelligence in wearable systems, as well as the ability and skills to apply the knowledge learned from the subject.

The materials submitted for this assessment must be the student's own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students' own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.

Student Study Effort Expected

Class contact:	
▪ Lecture	26 Hrs.
▪ Tutorial	12 Hrs.
Other student study effort:	
▪ Assignment/Project/Self-study	67 Hrs.
Total student study effort	105 Hrs.

Reading List and

Books

References	Steve Garner, (1991), *Human Factors*, Oxford University Press M. R. Lehto, S.J. Landry (2013), *Introduction to Human Factors and Ergonomics for Engineers*, Second Edition, CRC Press. J. Lazar, J. Feng, H. Hochheiser (2017), *Research Methods in Human-Computer Interaction*, Second Edition, Morgan Kaufmann G. J. Kim (2015), *Human–Computer Interaction: Fundamentals and Practice*, First Edition, Auerbach Publications A. Dix, J. Finlay, G. D. Abowd, R. Beale (2004), *Human–Computer Interaction*, Third Edition, Pearson Education Limited S. Russell, P. Norvig (2020), *Artificial Intelligence: A Modern Approach*, Fourth Edition, Pearson J. Kaplan (2016), *Artificial Intelligence: What Everyone Needs to Know*, Oxford University Press I. Goodfellow, Y. Bengio, A. Courville (2016), *Deep Learning*, The MIT Press E. Sazonov, M. R. Neuman (2014), *Wearable Sensors: Fundamentals, Implementation and Applications*, First Edition, Academic Press G. Fortino, R. Gravina, S. Galzarano (2018), *Wearable Computing: From Modeling to Implementation of Wearable Systems based on Body Sensor Networks*, First Edition, Wiley–IEEE Press. **Journals** Human Factors: The Journal of the Human Factors and Ergonomics Society (SAGE) IEEE Transactions on Human-Machine Systems (IEEE) Transactions on Interactive Intelligent Systems (ACM) IEEE Sensors Journal (IEEE) IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE) IEEE Transactions on Affective Computing (IEEE)

Note 1: Intended Learning Outcomes
Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus
The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology
This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method
This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.